RYAN O'DONNELL'S MATRIX CONJECTURE

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Abstract. This note records a statement refuted in the *Count ex Machina* archive: each is quoted as posed and answered by an explicit witness, and the archive carries a certificate that recomputes the refutation by exact arithmetic.

Summary: Every absolute c fails: an exact real-symmetric density-matrix family has ratio greater than $(\log_2 n)/32$. *Certificate:* An exact algebraic Givens-rotation family with a rational dyadic lower bound, accompanied by a standard-library exact-arithmetic verifier.

Posed by R. O'Donnell and J. Wright (2015); counterexample found by GPT-5.6 Pro (2026-08); formalized by GPT-5.6 Pro; audited by GPT-5.6 Pro; contributed by S. Sra.

For a Hermitian matrix X write $\|X\|_1 = \operatorname{tr}|X|$ for its trace norm. For $R \in \mathbb{C}^{n\times n}$, let $D = \operatorname{Diag}(R) = \operatorname{diag}(d_1, \ldots, d_n)$ be its diagonal part, and let $\Lambda(R)$ be the diagonal matrix carrying the eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$ in sorted order, so that

$$\epsilon := \|D - \Lambda(R)\|_1 = \sum\nolimits_i |\lambda_i - d_{ii}|.$$

The source states its conclusion asymptotically, as $\|R - D\|_1 = O(\sqrt{\epsilon})$; an absolute constant in that $O(\cdot)$ is exactly an absolute c with $\|R - D\|_1^2 \leq c\,\epsilon$, which is the form refuted below.

Problem 1 (As posed, paraphrased: R. O'Donnell, answering a question of Nengkun Yu on MathOverflow (O'Donnell, 2015), reports that he and J. Wright had been considering the problem in connection with quantum tomography, and closes by conjecturing the unit-trace case. The same conjecture was put to S. Sra by O'Donnell in personal communication.)**.** The question answered was whether a positive semidefinite M with $\|D(M) - \Lambda(M)\|_1 \leq \epsilon$ must lie within 2ϵ of conjectures the normalized case:

> "If you restrict M to have trace 1, then we felt the desired result was true but with a weaker conclusion of $O(\sqrt{\epsilon})$ rather than 2ϵ."

Written out, and with the implied constant absolute: for every unit-trace positive semidefinite R with eigenvalues λ_i and diagonal entries d_{ii},

$$\|R - \operatorname{Diag}(R)\|_1^2 \leq c \sum\nolimits_i |\lambda_i - d_{ii}|$$

for some absolute constant c independent of n.

Theorem 2 (**[Disproof]**: O'Donnell's matrix conjecture)**.** *For every $C > 0$ there exist an integer n and a PSD matrix $R \in \mathbb{R}^{n\times n}$ with $\operatorname{tr} R = 1$ such that, writing $D = \operatorname{Diag}(R) = \operatorname{diag}(d_1, \ldots, d_n)$ and listing the eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$,*

$$\frac{\|R - D\|_1^2}{\sum_{i=1}^n |\lambda_i - d_i|} > C.$$

Moreover, both (λ_i) and (d_i) are strictly decreasing.

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From the *Count ex Machina* archive of counterexamples, https://github.com/suvrit/count-ex-machina, whose companion paper is arXiv:2608.29595.

Proof. Fix an integer $m \geq 2$ and put $n = 2^m$, $q = n$, and $H = H_{n-1} = \sum_{i=1}^{n-1} \frac{1}{i}$. Then, set $B = 1 + q^2 H$, $\delta = \frac{1}{2B}$, and $\eta = \frac{1}{2n}$, and define positive spectral gaps $g_1 = \delta(1+q^2)$, $g_i = \delta q^2/i^2$ for $(2 \leq i \leq n-1)$. Then set the eigenvalues as

$$\lambda_n = \eta, \qquad \lambda_i = \eta + \sum_{k=i}^{n-1} g_k \quad (1 \leq i \leq n-1).$$

Clearly, $\lambda_1 > \cdots > \lambda_n > 0$, and $\sum_{i=1}^n \lambda_i = n\eta + \sum_{i=1}^{n-1} i g_i = \frac{1}{2} + \delta\left(1 + q^2 + q^2 \sum_{i=2}^{n-1} \frac{1}{i}\right) = \frac{1}{2} + \delta B = 1$.

Now for $1 \leq i \leq n-1$, put

$$r_i = \frac{q}{i}, \qquad c_i = \frac{q}{\sqrt{q^2 + i^2}}, \qquad s_i = \frac{i}{\sqrt{q^2 + i^2}}.$$

Hence $c_i^2 + s_i^2 = 1$ and $c_i/s_i = r_i$. Let G_i be the identity except for the following orthogonal block in coordinates $i, i+1$:

$$\begin{pmatrix} c_i & -s_i \\ s_i & c_i \end{pmatrix}.$$

Starting from $R^{(0)} = \Lambda$, define successively

$$R^{(i)} = G_i R^{(i-1)} G_i^{\mathsf{T}}, \qquad 1 \leq i \leq n-1,$$

and let $R = R^{(n-1)}$. This is an exact algebraic matrix. Since it is orthogonally similar to Λ, it is real symmetric, positive semidefinite, has trace one, and has eigenvalues (λ_i).

We next compute its diagonal exactly. Before the first rotation, the active diagonal gap is

$$\lambda_1 - \lambda_2 = g_1 = \delta(1 + r_1^2).$$

Before rotation G_i for $i \geq 2$, coordinate i has diagonal entry $\lambda_i + \delta$, coordinate $i+1$ has diagonal entry λ_{i+1}, and coordinate $i+1$ has not previously been touched. Thus the active 2×2 principal block is diagonal and its diagonal gap is

$$(\lambda_i + \delta) - \lambda_{i+1} = g_i + \delta = \delta(1 + r_i^2).$$

In both cases, $s_i^2\, \delta(1 + r_i^2) = \delta$. Consequently, G_i transfers exactly δ from diagonal position i to position $i+1$. Induction gives

$$d_1 = \lambda_1 - \delta, \qquad d_i = \lambda_i \quad (2 \leq i \leq n-1), \qquad d_n = \lambda_n + \delta.$$

These entries are strictly decreasing: at the first edge the remaining gap is $g_1 - \delta = \delta q^2 > 0$, at every interior edge it is $g_i > 0$, and at the last edge it is

$$g_{n-1} - \delta = \delta\left(\frac{q^2}{(n-1)^2} - 1\right) > 0$$

because $q = n$. Therefore $\sum_{i=1}^n |\lambda_i - d_i| = 2\delta$. The same rotation calculation gives useful off-diagonal entries. At step i, the newly created $(i, i+1)$ entry has magnitude $c_i s_i\, \delta(1 + r_i^2) = \delta r_i$. For $i \leq n-2$, the next rotation multiplies this entry by c_{i+1}; all later rotations leave it unchanged. Hence

$$|R_{i,i+1}| = \begin{cases} \delta r_i c_{i+1}, & 1 \leq i \leq n-2, \\ \delta r_{n-1}, & i = n-1. \end{cases}$$

Let $A = R - D$. Pinch A to the disjoint coordinate blocks $(1,2), (3,4), \ldots, (n-1, n)$. Trace norm is contractive under pinching, and each retained zero-diagonal 2×2 block has trace norm twice the magnitude of its off-diagonal entry. If

$$O = \sum_{\substack{1 \leq i \leq n-1 \\ i \text{ odd}}} \frac{1}{i},$$

then $q = n$ implies $c_j^2 = \frac{q^2}{q^2+j^2} > \frac{1}{2}$ $\quad (1 \le j \le n-1)$. Using the multiplier $1 > 1/\sqrt{2}$ for the final edge as well, we obtain

$$\|A\|_1 \ge 2 \sum_{\substack{1 \le i \le n-1 \\ i \text{ odd}}} |R_{i,i+1}| > \sqrt{2}\,\delta q O.$$

It follows that

$$\frac{\|R-D\|_1^2}{\sum_i |\lambda_i - d_i|} > \frac{2\delta^2 q^2 O^2}{2\delta} = \delta q^2 O^2 = \frac{q^2 O^2}{2(1+q^2 H)} \quad > \quad \frac{O^2}{2(H+1)}.$$

It remains only to estimate the two harmonic sums. In each dyadic block $[2^k, 2^{k+1})$, for $1 \le k \le m-1$, there are 2^{k-1} odd integers, each contributing more than $2^{-(k+1)}$. Therefore, $O > 1 + \frac{m-1}{4} = \frac{m+3}{4}$, and also $H_{n-1} < 1 + \log(n-1) < m+1$. Consequently,

$$\frac{\|R-D\|_1^2}{\sum_i |\lambda_i - d_i|} > \frac{(m+3)^2}{32(m+2)} > \frac{m}{32}.$$

Given any real $C > 0$, choose an integer $m > 32C$. The resulting exact matrix has ratio greater than C, proving that no dimension-independent constant can exist. □

Code and certificate

The case is the directory `counterexamples/odonnell-matrix-conjecture/` of the archive, at commit `73b416e`. Its `verify.py` recomputes every refutation in this note using exact arithmetic or rigorous interval enclosures, never floating-point evidence alone; it raises an error if any check fails and otherwise writes the certificate below. To rerun it:

```
git clone https://github.com/suvrit/count-ex-machina
cd count-ex-machina/counterexamples/odonnell-matrix-conjecture
python verify.py
```

- Theorem 2 (result `odonnell-matrix-conjecture`): exact arithmetic.
- `artifacts/certificate.json`: Exact algebraic family parameters, a symbolic unboundedness certificate, and a finite exact-arithmetic audit.

AI usage

The credit line under the title records, for each statement, who posed it, who found its counterexample and when, and who formalized and audited it; the counterexample here was found by GPT-5.6 Pro (2026-08). This note was assembled mechanically from the case's source in the archive by `tools/notes.py`, written with Claude Code; its mathematics is exactly that of the archive at the commit cited above.

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References

Ryan O'Donnell. Answer to "almost diagonal Positive semidefinite Matrix". MathOverflow answer 212759, August 2015. URL https://mathoverflow.net/a/212759. Answering question 212711 of Nengkun Yu; the conjecture is attributed there to J. Wright and the author, who record having also put it to S. Sra privately.